Land Titling, Gender, and Labor Mobility in China

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Abstract

We examine gender differences in the sectoral reallocation effects of land titling in rural China. Using the 2013–2019 China Household Finance Survey and village-level titling completion years, we exploit the reform's staggered rollout in a difference-in-differences design. Titling increases men's non-agricultural employment by 4.5 percentage points but produces no detectable aggregate response among women. We assess explanations involving household responsibilities, local job opportunities, and mobility costs. These factors help explain the pattern of men's reallocation out of agriculture, but do not account for women's lack of an aggregate response. We then explore household risk management as a complementary explanation. The joint employment, income, and consumption patterns are consistent with households expanding higher-return work while limiting changes in women's overall sectoral allocation, thereby stabilizing household income against potential fluctuations.

Keywords: Land Titling; Gender; Labor Mobility; Non-Agricultural Employment
JEL Codes: J16; J61; O13; Q15; D13

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1 Introduction

The movement of labor from agriculture to non-agricultural activities is central to structural transformation. Insecure land rights can impede this process when rural households fear that reducing cultivation or renting out their land may weaken their claims. By strengthening tenure security, land titling loosens the link between land retention and personal cultivation, thereby facilitating non-agricultural employment and labor mobility (Bu and Liao, 2022; De Janvry et al., 2015; Do and Iyer, 2008; Liu et al., 2023). However, because land rights are secured at the household level while the constraints and opportunities that shape labor supply vary across household members (Cao et al., 2026; Cortés and Pan, 2023; Liu et al., 2025), the same reform may produce different responses among women and men. Existing studies document the overall employment effects but say less about how the resulting labor adjustment is distributed within households. In this study, we examine whether land titling affects women's and men's employment differently and explore the possible causes of the gender differences.

We study China's nationwide land-titling program after 2014 by combining the 2013, 2015, 2017, and 2019 waves of the China Household Finance Survey (CHFS) with village-level titling completion years. Completion varied across villages within counties, allowing us to estimate a staggered difference-in-differences model with household and county-by-year fixed effects. These fixed effects absorb time-invariant characteristics of households and their villages, as well as annual shocks common to villages within the same county. After data cleaning and sample construction, the final sample contains 69,829 individual-year observations and 24,680 household-year observations across 1,568 villages. We measure employment using respondents' primary occupations and examine household members' participation in non-agricultural employment, with particular attention to gender differences.

The baseline results show that land titling increases non-agricultural employment by 2.4 percentage points, equivalent to 6 percent of the sample mean. The gender-specific estimates show a 4.5-percentage-point increase for men but no detectable change among women. This pattern indicates that land titling increases non-agricultural employment overall, consistent with earlier evidence on off-farm work and labor mobility (Bu and Liao, 2022; De Janvry et al., 2015; Do and Iyer, 2008). In our setting, however, the increase is driven by men. We further estimate a pooled specification that interacts land titling with a female indicator. The interaction estimate shows that the female response is 2.7 percentage points smaller than the male response. This difference documents a gender-heterogeneous effect of land titling on non-agricultural employment.

We assess the identification strategy and robustness of this gender pattern in several ways. The event-study estimates show no detectable differential trends before titling completion. After completion, the estimates remain close to zero for women, whereas those for men become positive

and generally increase over time. Observed baseline village characteristics do not predict the timing of titling completion detectably, reducing concerns that the rollout was systematically related to these characteristics. Alternative estimators robust to heterogeneous treatment effects in staggered DID yield similar estimates. The central pattern remains unchanged when we replace household fixed effects with individual fixed effects, account for differential trends associated with baseline village characteristics, and exclude early pilot areas. Together, these exercises show that the gender difference in employment responses persists after land titling.

Building on research showing that care responsibilities, household specialization, and access to work shape labor allocation (Becker, 1985; Cao et al., 2026; Cortés and Pan, 2023; Le Barbanchon et al., 2021; Liu et al., 2025), we examine heterogeneity along these household and local labor-market dimensions to explore the causes for the gender difference. The estimates show that men's non-agricultural employment rises mainly when care demands are lighter, industrial employment is more accessible, and villages are closer to county centers. The male response also varies with spouses' relative education, while migration itself does not change detectably. In contrast, the female estimates remain close to zero across these settings. Observable household and labor-market conditions therefore help identify where men's employment responds, but they do not account for the absence of an aggregate response among women.

To interpret this remaining gender difference, we examine household labor allocation through the lens of risk management. Households may pursue higher returns by assigning a larger sectoral adjustment to one spouse while limiting their overall exposure to employment and income risk. Consistent with this interpretation, men's non-agricultural employment rises, whereas women's aggregate sectoral allocation remains stable; among employed women, the composition of work shifts from flexible to wage employment. These employment changes coincide with higher non-agricultural income but no detectable increase in current consumption. The joint pattern is consistent with households undertaking the larger sectoral adjustment through men while maintaining a comparatively stable employment base among women (Giles and Yoo, 2007; Morduch, 1995).

The paper makes three contributions. First, we bring an intra-household gender perspective to the literature on land rights and labor allocation. Existing research shows that stronger land rights affect non-farm labor supply and enterprise activity (Bu and Liao, 2022; Do and Iyer, 2008), migration (De Janvry et al., 2015; Valsecchi, 2014), and agricultural productivity through the reallocation of land and human capital (Liu et al., 2023). This literature establishes that tenure security can release labor from agriculture, but it focuses mainly on household-level or aggregate labor reallocation. We instead examine how this adjustment is distributed among household members and show that the average increase in non-agricultural employment masks a gender-asymmetric response. We also measure reform exposure using village-level titling completion

years rather than the county-level start dates used by Bu and Liao (2022) or completion dates used by Liu et al. (2023). This finer timing reduces treatment misclassification arising from within-county rollout and allows us to compare villages within the same county and year.

Second, we connect land-induced labor reallocation to the broader literature on gender differences in labor supply. Prior work shows that care responsibilities and household specialization shape women's and men's employment differently (Becker, 1985; Cao et al., 2026; Cortés and Pan, 2023; Wang, 2023), while job opportunities and mobility costs affect access to work (Le Barbanchon et al., 2021; Liu et al., 2025; Wang et al., 2024). In the land context, Shi et al. (2024) find that stronger land-leasing rights raise off-farm employment more for men than for women. We show where this difference emerges: the male response is concentrated where care burdens are lighter and employment opportunities are more accessible, whereas the corresponding female estimates remain close to zero. These findings show that relaxing a land-related constraint does not produce a common employment response within the household; family responsibilities and labor-market access continue to shape the observed adjustment.

Third, we contribute to research on joint household labor supply and risk sharing, which treats spouses' employment choices as interdependent responses to household opportunities and risks (Halla et al., 2020; Lundberg, 1985; Shore, 2010). In the land context, Adamopoulos et al. (2024) model the household choice of who farms and who migrates. We add empirical evidence by examining spouses' sectoral and job-type responses jointly with household income and consumption after titling. The joint pattern is consistent with households pursuing higher returns through men's sectoral reallocation while maintaining a comparatively stable employment base among women (Giles and Yoo, 2007; Morduch, 1995). This perspective broadens the analysis from whether labor leaves agriculture to how households allocate adjustment across spouses while balancing returns and risk.

The remainder of the paper proceeds as follows. Section 2 describes China's land-titling program and places gender differences in rural employment and migration in context. Section 3 introduces the data and empirical strategy. Section 4 presents the baseline estimates and robustness checks. Section 5 reports heterogeneity by household structure, care burden, and local labor-market conditions. Section 6 develops the household risk-management interpretation using employment, income, and consumption evidence. Section 7 concludes.

2 Institutional Setting and Labor-Market Context

2.1 Land Titling Reform in China

In China, village collectives own rural land and allocate agricultural plots to households. The Household Responsibility System (HRS), introduced in the early 1980s, granted village members use rights to land under an egalitarian allocation system and substantially increased grain production (Lin, 1992; McMillan et al., 1989). As urbanization and industrialization advanced, agriculture's share of GDP declined (Schultz, 1951), while rural-to-urban migration and the movement of labor from agriculture to non-agricultural activities expanded, particularly after China's accession to the WTO in 2001 (Erten and Leight, 2021; Facchini et al., 2019).

Initially, village members received partly "use-based" land rights under 15-year contracts.[1] Households were generally required to cultivate their allocated plots themselves. Before the 2003 Rural Land Contracting Law, their rights to rent land in or out were not clearly codified, and agricultural land could not be bought or sold. These restrictions contributed to the misallocation and inefficient use of agricultural land (Adamopoulos et al., 2024; Chen et al., 2026). Although the 2003 law granted farmers rights to transfer land, villages could still reclaim and reallocate plots left uncultivated, leaving residual property rights ambiguous. This "use-it-or-lose-it" arrangement gave households a reason to keep labor in agriculture even when non-agricultural work offered higher returns (Adamopoulos et al., 2024; Brandt et al., 2002).

The central government and the Ministry of Agriculture (MOA) began piloting land titling in selected counties and cities in 2009 (Liu et al., 2023), and large-scale nationwide implementation began in 2014. By the end of 2018, the program had been carried out in 2,838 county-level units (Ministry of Agriculture and Rural Affairs of China, 2019). Using CHFS data, Appendix Figure A1 documents the village-level rollout of land titling, which broadly matches the national implementation schedule described above. We find that village-level implementation began primarily in 2014 and accelerated over subsequent years: more than 80 percent of villages with reported completion years had finished titling by 2018, and all had done so by 2019.

Because the program was implemented through a top-down administrative process, national rollout did not translate immediately into village-level completion. Appendix Figure A2 depicts the implementation chain from the central government to the village level. In practice, the reform followed three phases[2]: land measurement, registration, and certification. In the first phase, a

[1] In the early 1980s, only residents registered in the village were eligible to receive a land allocation. Rural land remained collectively owned, and households received contractual use rights rather than ownership. These rights were initially granted for 15 years. In the 1990s, the Land Management Law (LML) extended household land-use rights for another 30 years to reduce the tenure insecurity created by land readjustments (Adamopoulos et al., 2024; Liu et al., 2023).

[2] Official guidance for the confirmation, registration, and certification process is provided in the Ministry of

specialized team produced a geocoded map at a 1:2,000 scale, detailing land types, serial numbers, and plot areas for each village. In the second phase, village organizations verified land-use rights using the mapped data. In the final phase, county-level land-management offices registered land operator–parcel pairs, performed error checks, and issued land certificates. Completing these steps usually took about one and a half years (Liu et al., 2023), and villages within the same county could therefore complete titling at different times.

Although land titling did not alter collective ownership, it strengthened households' contractual rights by formally recording plot boundaries and issuing legally recognized certificates. These certificates reduced ambiguity over landholdings and made arbitrary land reallocations and uncompensated expropriation more difficult.

2.2 Gender Differences in Employment in Rural China

As industrialization and urbanization drew rural labor out of agriculture, women and men entered the expanding non-agricultural economy at different rates. Between 1990 and 2015, labor force participation declined among rural adults aged 20–40, and the decline was steeper among women than among men. Over the same period, a growing share of rural women reported themselves as housewives (Wang, 2023). The gender gap therefore emerged at two related margins: whether individuals remained in the labor force and, conditional on working, whether they moved out of agriculture.

The difference in sectoral allocation was substantial. In 2010, 53.1 percent of male rural hukou holders in the labor force worked in non-agricultural sectors, compared with 43.2 percent of their female counterparts. This gap widened between 2000 and 2010 and was concentrated among married workers; single women were no less likely than single men to work outside agriculture (Cao et al., 2026). The same pattern was present within rural couples. Among couples whose spouses worked in different sectors, the husband was more often in non-agricultural employment and the wife in agriculture than the reverse. Rural labor reallocation therefore remained closely connected to the division of labor within households.

The composition of non-agricultural employment also differed by gender. Rural men were more likely to work in construction and other physically demanding occupations and to migrate for employment, while women were more represented in local than in outgoing migrant employment (National Bureau of Statistics of China, 2020; Shi et al., 2024). Women also received

Agriculture et al., *Opinions on Properly Conducting the Confirmation, Registration, and Certification of Rural Land Contractual Management Rights*, Nong Jing Fa [2015] No. 2, available at https://fgs.moa.gov.cn/flfg/202011/t20201124_6356933.htm, and the Ministry of Agriculture and State Archives Administration, *Measures for the Archival Management of the Confirmation, Registration, and Certification of Rural Land Contractual Management Rights*, Nong Jing Fa [2014] No. 12, available at https://www.moa.gov.cn/nybgb/2014/shier/201712/P020180104778141801165.pdf. These documents set out the confirmation, registration, certification, and archival procedures underlying the rollout.

lower earnings from non-agricultural work (Cao et al., 2026; Shi et al., 2024). Gender differences in rural employment thus extended beyond participation to sector, occupation, location, and earnings. Against this background, a reform that made it easier to leave agriculture could increase non-agricultural employment without producing the same adjustment among women and men.

3 Data and Empirical Strategy

3.1 Data

We use household- and individual-level panel data from the China Household Finance Survey (CHFS). The CHFS covers 29 provincial-level administrative units in mainland China and is representative nationally and at the provincial level. Alongside detailed information on household finances, the CHFS records rich individual information, including respondents' employment, education, demographic characteristics, and relationships to other household members (Gan et al., 2016). We combine the 2013, 2015, 2017, and 2019 waves to construct an unbalanced panel and merge the individual and household records with the corresponding village dataset.[3] We focus on rural households and individuals aged 17–60 who were no longer enrolled in education when their village first entered the treated period. The resulting sample contains 69,829 individual-year observations and 24,680 household-year observations across 1,568 villages.

Our primary outcome is an indicator for non-agricultural employment. It equals one when the respondent's primary occupation is non-agricultural and zero otherwise. The village questionnaire records the year in which each village completed land titling. We define a village as treated in a given survey year if titling was completed before that survey year. Thus, a village that completed titling in a given calendar year first enters the treated period in the subsequent survey wave. This village-level timing allows us to assign treatment status more precisely than the county-level timing measures commonly used in prior studies (Bu and Liao, 2022; Liu et al., 2023).[4] Implementation was staggered across villages within a county, so county-level start or completion dates would misclassify many villages. Appendix Figure A3 documents this within-county variation by plotting the difference between village and county completion years. The figure shows that same-year completion is the most common outcome, but many villages finished either before or after the county completion date. Village-level timing reduces this measurement error and allows us to compare villages within the same county and year.

[3]The CHFS does not observe every sampled village or household in every survey wave, so the set of villages with an observed completion year varies across the early period. We exclude villages without a reported completion year or with a reported completion year before 2008. When the questionnaire reports multiple completion years for a village, we use the most recent report.

[4]For example, Bu and Liao (2022) use the county start year and Liu et al. (2023) use the county completion year. Neither measure captures the year in which a particular village completed titling.

Table 1 displays summary statistics for our primary analysis samples.[5] Panel A describes the 1,568 villages. The average village has 1,873 residents and 441 members of the village labor force working outside their place of household registration or usual household residence, equivalent to about 24 percent of the village population. In 95 percent of villages, at least one large surname group (*Daxing*) accounts for more than 10 percent of the population, indicating that kinship-based organization is common. Thirty-seven percent of villages implement the *yijiantiao* ("one-shoulder") arrangement, in which one person serves as both village Party secretary and village committee director.[6] The average village covers 95,810 *mu* (about 6,387 hectares) and lies 31 kilometers from the county center.

Panel B of Table 1 describes the 24,680 household-year observations. The typical household has 4.28 members and reports annual per capita income of 13,140 yuan, total consumption of 12,300 yuan, savings of 4,480 yuan, assets of 104,680 yuan, and debt of 11,200 yuan. Panel C describes the 69,829 individual-year observations. The average respondent is 40 years old and has 8.7 years of schooling. Seventy-seven percent are employed. Based on respondents' primary occupations, 35 percent work in agriculture and 40 percent in non-agricultural sectors.[7] Wage employment and flexible employment each account for about 15–20 percent of the full sample, while business employment, including self-employment, accounts for 5 percent. We retain these job-type labels throughout the paper to match the CHFS employment categories.

[Insert Table 1 about here]

3.2 Empirical Strategy

We estimate the effects of land titling on individual labor outcomes using the following staggered difference-in-differences specification:

$$y_{ihvct} = \beta \text{Titling}_{vt} + X_{ihvct}\rho + \eta_h + \delta_{ct} + \varepsilon_{ihvct} \tag{1}$$

where y_{ihvct} denotes the labor-market outcome of individual i in household h, village v, county c, and year t. The indicator Titling_{vt} equals one beginning in the first survey year after land titling completion and in all subsequent survey years. The coefficient β compares changes in labor outcomes for individuals in villages that have completed titling with contemporaneous changes for individuals in not-yet-treated villages within the same county. Household fixed effects, η_h,

[5] We winsorize income and other continuous economic measures at the 1st and 99th percentiles.

[6] The *yijiantiao* arrangement consolidates Party and administrative leadership and may proxy for local state capacity or the concentration of village authority (Marie et al., 2024).

[7] Employment variables refer to the respondent's primary occupation. When a respondent holds more than one job, the CHFS records the occupation receiving the most work time; the reported sector and job-type shares therefore exclude secondary activities.

absorb time-invariant observed and unobserved characteristics shared by members of the same household.[8] County-by-year fixed effects, δ_{ct}, absorb contemporaneous labor-market conditions, policies, and other shocks common to all villages in a county. The vector X_{ihvct} contains age, gender, and years of schooling. We cluster standard errors at the village level.

We first estimate equation (1) in the pooled sample and then separately for women and men. To test whether the treatment response differs by gender, we augment the pooled specification with the interaction between the land-titling indicator and the female indicator. The interaction coefficient measures the female treatment response relative to the male response, conditional on the common controls and fixed effects.

The identification strategy requires that, absent titling, villages completing the reform at different times would have followed similar employment trends, conditional on the fixed effects and controls. We examine observed pre-titling dynamics and the evolution of individuals' responses using the following event-study specification:

$$y_{ihvct} = \sum_{\substack{k=-4 \\ k\neq -1}}^{4} \beta_k D_{vt}^{k} + X_{ihvct}\rho + \eta_h + \delta_{ct} + \varepsilon_{ihvct} \tag{2}$$

where the control variables are the same as in equation (1). We define the event-time index as $k = t - T_v - 1$, where T_v is the village's land-titling completion year. Thus, $k = -1$ corresponds to the completion year and is the omitted category, while $k = 0$ denotes the calendar year immediately after completion. The indicators for $k \leq -4$ and $k \geq 4$ are binned at -4 and 4, respectively. We estimate equation (2) separately for women and men. The coefficients for $k \leq -2$ describe employment differences before completion, while those for $k \geq 0$ trace the post-reform response, each relative to the completion year ($k = -1$).

4 Employment Responses by Gender

4.1 Baseline Results

Table 2 reports the baseline results estimated using equation (1). The dependent variable is an indicator for non-agricultural employment. Column (1) shows that land titling increases the probability of non-agricultural employment by 2.4 percentage points. Relative to the sample mean of 40 percent, this is an increase of about 6 percent, consistent with evidence that secure land rights facilitate movement away from agriculture (De Janvry et al., 2015; Liu et al., 2023; Valsecchi, 2014). The pooled estimate, however, masks different responses by gender. Columns

[8] As a robustness check, we replace household fixed effects with individual fixed effects.

(2) and (3) estimate the same equation separately for women and men. The results indicate a 4.5-percentage-point increase for men, whereas the estimated effect for women is -0.2 percentage points and statistically indistinguishable from zero. This pattern points to a gender-asymmetric employment response. To formally test this difference, column (4) estimates a pooled interaction specification. The treatment response for women is 2.7 percentage points smaller than that for men.

[Insert Table 2 about here]

Conventional two-way fixed-effects estimates may be biased under staggered adoption and heterogeneous treatment effects (Baker et al., 2022). Appendix Table B1 therefore reports average post-treatment effects from the interaction-weighted estimator of Sun and Abraham (2021) and the stacked DID approach of Cengiz et al. (2019), alongside the TWFE estimate. The alternative estimates remain close to zero for women (-0.8 and -2.0 percentage points) and positive for men (5.7 and 9.1 percentage points). Because these entries average the dynamic post-treatment coefficients rather than estimate a single treatment indicator, they complement the static TWFE estimate and reproduce the baseline difference by gender.

Consequently, the pooled estimate should not be interpreted as a uniform response across household members. Together with the gender-specific estimates, the negative interaction term shows that the average employment increase in column (1) is concentrated among men. Although land titling secures property rights at the household level, untying land retention from personal cultivation produces different labor-reallocation responses among women and men.

4.2 Identification Diagnostics

We examine two diagnostics relevant to our staggered difference-in-differences design. The first is whether villages display differential employment dynamics before titling; the second is whether observed village characteristics predict the timing of titling completion.

Pre-Treatment Dynamics. The difference-in-differences design assumes that, absent titling, non-agricultural employment would have evolved similarly in villages completing the reform at different times. Observed pre-treatment coefficients cannot establish this counterfactual assumption, but they can reveal differential outcome dynamics before completion. Because conventional two-way fixed-effects estimates can be biased when treatment effects vary across cohorts or event time (Goodman-Bacon, 2021), we also employ the methods proposed by Cengiz et al. (2019) and Sun and Abraham (2021).[9] Figure 1 plots the three sets of dynamic estimates separately for women

[9]The stacked approach builds a separate dataset for each treatment cohort using never- or not-yet-treated villages as controls and then combines the cohort-specific samples. The interaction-weighted approach estimates cohort-by-event-time effects and aggregates them using cohort shares.

and men. Before completion ($k \leq -2$), the coefficients are small and statistically indistinguishable from zero in both panels. We therefore find no detectable systematic differences in pre-completion trends over the observed event window. This evidence is consistent with, but does not prove, the identifying assumption. After completion ($k \geq 0$), the estimates for women remain close to zero, whereas those for men become positive and generally grow over event time. The post-completion pattern is similar across the three estimators and, together with the pooled interaction estimate, reinforces the evidence of gender-differentiated employment responses.

[Insert Figure 1 about here]

The Timing of Land Titling. To assess the timing of treatment, we examine whether village-level titling completion years are associated with predetermined characteristics.[10] Appendix Figure A4 reports separate regressions of the titling-completion year on village population, the number of out-migrants, the presence of a large surname group (*Daxing*), the *yijiantiao* ("one-shoulder") arrangement, village area, distance to the county center, the number of roads, and farmland per capita. The estimates are small and do not differ detectably from zero. Thus, the observed baseline characteristics do not predict completion timing detectably.

4.3 Robustness Checks

We assess whether the baseline gender pattern depends on four features of the empirical design: the inclusion of early pilot areas, the choice between household and individual fixed effects, differential trends associated with baseline village characteristics, and sample composition by pre-reform occupation. Table 3 reports these checks.

[Insert Table 3 about here]

Excluding Early Pilots. Early pilot sites were concentrated in Sichuan and may have differed from the standardized nationwide rollout. During the experimentation phase, the pilot program generated practical knowledge that informed the national land-titling policy (Deininger et al., 2020; Liu et al., 2023). Columns (1)–(3) in Panel A exclude Sichuan Province. The gender-specific estimates remain close to the baseline results, and the pooled female interaction remains −2.8 percentage points.

Individual Fixed Effects. Household fixed effects do not absorb time-invariant characteristics that differ among members of the same household.[11] Columns (4)–(6) in Panel A therefore replace

[10] Specifically, the dependent variable is the year of titling completion, and the explanatory variables are baseline village characteristics. Standard errors are clustered at the village level.

[11] Since high-dimensional fixed effects may exacerbate measurement error bias and increase the likelihood of false positives (Huang et al., 2024; Jennings et al., 2024), we control for household fixed effects instead of individual fixed effects in the baseline analyses.

them with individual fixed effects. The results show that land titling raises men's non-agricultural employment by 2.9 percentage points, while the estimate for women remains near zero. The pooled female interaction is −3.2 percentage points, so the baseline pattern persists after absorbing time-invariant individual characteristics.

More Controls. Local governments may have completed titling earlier in villages already following different demographic, migration, or agricultural trajectories. Columns (1)–(3) in Panel B interact year indicators with five village characteristics measured in the baseline survey wave:[12] out-migration, the presence of a large surname group (*Daxing*), distance to the county center, registered (*hukou*) population, and farmland per capita. This specification leaves the baseline gender pattern largely unchanged.

Pre-Reform Farmers. If titling releases labor from cultivation, the response should be most visible among people who previously worked in agriculture. Columns (4)–(6) in Panel B restrict the sample to individuals whose primary occupation before village completion was in agriculture.[13] In this sample, titling raises men's non-agricultural employment by 11.6 percentage points, while the female estimate remains close to zero. The pooled female interaction is −9.6 percentage points. The larger male estimate among prior farmers is consistent with the reform operating through a labor-reallocation margin because this group had greater scope to move out of agriculture.

5 Probing Mechanisms Through Heterogeneity

The baseline results establish different aggregate and compositional responses for women and men. A natural next question is what accounts for this gender difference in employment responses. The literature emphasizes household care burdens, intra-household bargaining, and labor-market constraints. We assess their plausibility by examining how the estimated effects vary across household characteristics, local labor-market structures, and geographic mobility barriers. These subgroup comparisons do not identify a single mechanism. Instead, they show whether the gender pattern is concentrated in settings where particular constraints are likely to bind more tightly.

[12] "Baseline" denotes the first CHFS wave used in the analysis rather than a common pre-reform year. These characteristics are measured in 2013, and some village-level controls have fewer observations because some villages lack these measures. When the 2013 out-migrant count is missing, we use the corresponding 2015 report.

[13] This group accounts for about 35 percent of the full sample, matching the agricultural employment share in Table 1.

5.1 Household Structure

We first examine whether the employment response varies with family structure. The analysis focuses on domestic care responsibilities, marital status, and spouses' relative education, which may constrain women and men differently.

5.1.1 Family Care Burden

Childcare may limit a household's ability to move labor away from the village, and women often carry a larger share of this responsibility (Cortés and Pan, 2023; Maurer-Fazio et al., 2011; Sun et al., 2026). To examine the role of care burden, we restrict the analysis to married individuals and split the sample separately by the presence of children aged 15 or below and by the presence of older household members in poor health.[14] Panel A of Table 4 shows that titling significantly raises men's non-agricultural employment by 6.0 percentage points in households without young children; the estimate is smaller and imprecise when young children are present. Panel B shows a similar pattern: titling significantly raises men's employment by 4.6 percentage points in households without unhealthy older members but produces no detectable change among men living with such household members.

These sample splits capture distinct dimensions of domestic constraints. The child split aligns with theories of household specialization around childcare and home production (Becker, 1985). Older household members in poor health may instead create eldercare responsibilities. Related research shows that care constraints can increase unpaid work among female household members, not only mothers (Aguilar-Gomez et al., 2026). The female estimates remain close to zero across all care-burden categories. Lighter care demands therefore help locate the male response but do not generate a detectable aggregate response among women.

[Insert Table 4 about here]

Treating the two dimensions separately avoids assuming that co-resident older members only create care needs; older household members may also provide childcare (Bratti et al., 2018). Appendix Table B2 reports the full 2×2 split, categorizing households into four mutually exclusive groups based on the joint presence of young children and unhealthy older members. The estimates differ detectably from zero only for men facing the lightest observed care burden—those with neither young children nor unhealthy older members—whose non-agricultural employment rises by 8.2 percentage points. The female estimates remain statistically indistinguishable from zero across all four groups. This pattern reinforces that the male response is concentrated among

[14] The underlying health measure ranges from 1 (very good) to 5 (very poor). We define an older household member as unhealthy when the response is poor or very poor (4 or 5); the indicator equals zero for very good, good, or fair health (1–3).

households with lighter observed care demands, whereas no corresponding aggregate response emerges among women.

5.1.2 Marital Status and Education

Marriage creates scope for household specialization because spouses may coordinate their employment choices (Chiappori et al., 2002). Panel C of Table 4 examines whether the response to titling differs by marital status. The results show that land titling significantly raises non-agricultural employment by 3.6 percentage points among married men, while married women show no detectable aggregate response. The estimates for unmarried women and men are both imprecise due to their wide standard errors.[15] This pattern is compatible with household specialization once couples make labor decisions jointly (Becker, 1985); differing constraints on temporal flexibility can reinforce gender gaps in market work (Goldin, 2014).

Within-household bargaining power may affect the division of labor. We examine relative education because it may capture differences in outside options and influence over household decisions (Duflo, 2012; Li, 2026). A stronger educational position may increase the wife's influence over household decisions, while a weaker relative position may coincide with more traditional specialization. We split married couples by the wife's education relative to her husband's. We classify the wife's relative education as "High" when her schooling equals or exceeds her husband's and as "Low" otherwise.[16] Panel D of Table 4 shows small and imprecise responses for both spouses when the wife's education equals or exceeds her husband's. When the wife has less education, titling significantly raises the husband's non-agricultural employment by 8.4 percentage points, while the wife's estimate remains near zero. The directional pattern is compatible with a role for relative status in household labor allocation (Bertrand et al., 2015).

5.2 Local Labor-Market Opportunity

Two features of the local labor market may shape the observed gender difference: occupational segregation by gender and the overall availability of non-agricultural jobs. First, if occupational segregation restricts women's access to non-agricultural work, we would expect a stronger female employment response in counties with less segregation (Blau and Kahn, 2017; Kuhn and Shen, 2013). To test this, we follow Duncan and Duncan (1955) and Blau and Hendricks (1979) by

[15] The "unmarried" category combines never-married, divorced, and widowed individuals, who differ in household structure and caregiving circumstances. Each subgroup is also small.

[16] This measure is an imperfect proxy for bargaining power because spouses may sort on education. It may capture household matching and relative labor-market position in addition to bargaining power. We therefore interpret the estimates as heterogeneity by within-household relative status, not as a structural bargaining parameter.

constructing a county-level Duncan index using data from the 2010 Population Census of China.[17] This measure explicitly captures the extent to which women and men are differentially distributed across local non-agricultural occupations. Second, if the absence of an aggregate female response is driven by limited local job opportunities, women's employment should rise more prominently in areas with a higher concentration of industrial firms. We proxy for this local job availability using the number of industrial enterprises above the designated size, as reported in county statistical yearbooks.[18]

[Insert Table 5 about here]

Panel A of Table 5 splits counties by occupational segregation. In counties with above-median occupational segregation, titling significantly raises men's non-agricultural employment by 6.6 percentage points, while the female estimate is small and negative. In counties with below-median occupational segregation, both estimates are positive but do not differ detectably from zero. Panel B turns to job availability. In counties with more industrial firms, titling significantly raises men's non-agricultural employment by 5.6 percentage points, while the female estimate remains near zero. Greater local job availability therefore helps locate the male response but does not generate a detectable aggregate response among women.

5.3 Migration and Geographic Mobility

Because land titling lowers the cost of leaving the village, migration is a natural component of the labor-mobility response to titling (Adamopoulos et al., 2024; Chen et al., 2026; Ngai et al., 2019). Women may face greater commuting constraints than men because care responsibilities and social expectations can narrow women's geographic job search and increase their dependence on nearby work.[19] We therefore examine whether migration responds differently by gender.

We measure migration using an indicator equal to one if an individual leaves the county for more than three months and estimate this outcome using equation (1).[20] Panel A of Table 6 uses the baseline sample.

[17] The index for county c is $D_c = 1/2 \sum_{i=1}^{n} |f_{ci}/F_c - m_{ci}/M_c|$, where f_{ci} and m_{ci} are female and male employment in occupation i, and F_c and M_c are total female and male employment. The index ranges from zero to one, with higher values indicating greater occupational segregation. Because it is constructed from the 2010 census, it predates the nationwide titling program.

[18] Industrial enterprises above the designated size are firms with annual revenue from principal business of at least 20 million yuan. Counties with more such firms have a denser non-agricultural labor market, which may allow rural residents to change sectors without moving long distances.

[19] For example, women set shorter acceptable commutes during job searches in France (Le Barbanchon et al., 2021), and longer commuting times reduce married women's labor-force participation in China (Liu et al., 2025).

[20] This measure captures longer-distance migration but does not record moves within the county and therefore provides an incomplete measure of geographic mobility.

Titling increases the migration indicator by 1.5 percentage points for women and 2.0 percentage points for men, but neither estimate differs detectably from zero. We therefore find no direct evidence that land titling increases cross-county migration. Columns (3) and (4) add migration status to the non-agricultural-employment regression. Relative to the baseline estimate of 4.5 percentage points, the male titling coefficient declines to 3.9 percentage points but remains positive. The female coefficient changes from −0.2 to −0.7 percentage points and remains close to zero. The modest attenuation of the male coefficient is consistent with cross-county migration accounting for part of men's employment response, although this post-treatment control does not identify a causal mediation effect.

[Insert Table 6 about here]

Closely related to migration are commuting distance and travel time, which may shape employment opportunities differently by gender (Borghorst et al., 2024; Farré et al., 2023). We therefore conduct a subgroup analysis based on the village's distance to the county administrative center. Panel B compares villages by this distance—a direct measure of mobility cost.[21] The male estimate is 10.9 percentage points in villages closer to the county center and 3.8 percentage points in more distant villages. This directional pattern is consistent with lower mobility costs facilitating men's employment response. The female estimates do not differ detectably from zero in either group, including a negative estimate in nearby villages. Lower distance-related mobility costs therefore do not produce a detectable aggregate female response.

5.4 Summary

Taken together, the heterogeneity estimates in Tables 3, 4, 5, and 6 locate the male response in settings where domestic care demands are lighter, local industrial jobs are more abundant, and spatial mobility costs are lower. The subgroup comparisons do not establish that these conditions cause the response, but they show that the increase in men's non-agricultural employment is concentrated where labor reallocation is more feasible. The female estimates remain close to zero across the same settings. Observable household and labor-market conditions therefore help explain where men respond but do not account for the absence of an aggregate response among women.

The evidence presented thus far does not identify a single observable constraint that fully accounts for this contrast. This pattern resembles the findings of Shi et al. (2024), who also document a larger male than female off-farm employment response to a land-rights reform in

[21] "Close" villages lie below the sample median distance of approximately 31 kilometers.

rural China.[22] To better understand this gender contrast, the next section turns to household risk management and asks whether households combine stronger labor reallocation among men with a relatively stable overall employment allocation among women to limit exposure to income risk.

6 Household Risk Management

The labor reallocation documented above can also be viewed through the lens of household risk management. A large literature emphasizes that household members coordinate their labor supply decisions in response to income risk. Couples may adjust labor supply jointly following common shocks (Shore, 2010), while at the same time diversifying employment across household members to reduce exposure to adverse shocks. The classic "added worker effect" (Lundberg, 1985), whereby married women temporarily increase their labor supply following their husbands' unemployment, provides an early example of such intra-household adjustment.[23]

This perspective is particularly relevant in rural China, where agricultural and non-agricultural activities differ not only in expected returns but also in the risks they entail. By strengthening land property rights, land titling reduces the cost of reallocating labor away from agriculture and facilitates entry into higher-return non-agricultural activities (Adamopoulos et al., 2024; Liu et al., 2023). Yet a household in which both spouses adjust toward the same exposed activity may become more vulnerable to unemployment, wage arrears, and fluctuations in labor demand. These considerations generate a role for intra-household specialization: the household may concentrate the larger sectoral adjustment among one spouse while keeping the other's overall employment allocation relatively stable. In our setting, this arrangement would allow men to pursue higher-return non-agricultural work while the relative stability of women's overall sectoral allocation provides a more stable employment base.[24]

This framework yields two empirical implications. First, land titling should facilitate movement toward employment categories with higher expected earnings. Second, if households use labor allocation to manage income risk, the response need not involve similar sectoral changes for both

[22] The female baseline coefficient is statistically indistinguishable from zero, but failure to reject the null does not establish that the true effect is exactly zero. A formal claim that the effect is negligibly small would require an equivalence test or a pre-specified minimum-effect analysis (Gelman and Stern, 2006; Lakens, 2017).

[23] This interpretation relates to two strands of the literature. First, models of household labor portfolios emphasize that household members may diversify employment risks across occupations. For example, Parker (2008) show that stable employment by one household member can induce the spouse to enter higher-risk, higher-return occupations, while Li et al. (2026b) documents that having a spouse employed in the state-owned sector increases individual entrepreneurship. Second, collective household models treat the household as an integrated decision-making unit and incorporate income uncertainty into the allocation of labor and resources within the household (Chiappori, 1988, 1992).

[24] The hukou system and the costs of whole-family migration reinforce such arrangements. Rural households frequently adopt split-migration strategies in which one spouse migrates for work while the other remains in the village and continues agricultural production (Mu and van de Walle, 2011).

spouses, nor should higher income necessarily translate immediately into higher consumption. Instead, households may combine a stronger movement toward higher-return work by one spouse with a relatively stable overall employment allocation for the other, while retaining part of the resulting income gains rather than fully converting them into current consumption.

6.1 Expected Returns and Employment Reallocation

We first examine whether non-agricultural employment categories are associated with different earnings opportunities. Figure 2 plots the distribution of individual income by gender across three non-agricultural employment types—wage employment, business employment, and flexible employment—and reports the corresponding group means.[25]

The income distributions reveal the same ranking across employment types for women and men. Among men reporting positive income, mean income is 32,254 yuan in wage employment, 27,553 yuan in flexible employment, and 11,232 yuan in business employment. The corresponding means for women are 26,674 yuan, 20,163 yuan, and 5,088 yuan. Wage employment is therefore associated with the highest observed earnings for both genders, flexible employment occupies an intermediate position, and business employment has the lowest mean. Men also have higher mean income than women within each category. These descriptive comparisons do not identify the causal return to an employment type. Flexible employment, by definition, includes work without a formal labor contract and may therefore expose workers to greater job-loss risk. The data do not directly measure employment stability or unemployment risk, so this distinction remains an interpretation of the employment categories rather than an estimated risk differential.

[Insert Figure 2 about here]

Table 7 examines how land titling changes primary-employment composition among respondents who are currently employed. The four mutually exclusive categories are agriculture, wage employment, flexible employment, and business employment. Columns (1) and (2) show that men's agricultural-employment share declines by 4.7 percentage points, while women's agricultural share changes little. Columns (3) and (4) indicate that the wage-employment share rises by 2.9 percentage points for women and 2.8 percentage points for men. The accompanying margins differ by gender. For men, the increase in wage employment occurs alongside the decline in agricultural employment, while flexible and business employment show no detectable changes. For women, the flexible-employment share falls by 3.2 percentage points, while business employment changes

[25]Business employment refers to self-employment and entrepreneurial activities. Flexible employment refers to work without a formal employment contract, including freelance work and voluntary work. We do not report individual agricultural income because agricultural earnings are generally measured at the household level and cannot be reliably assigned to individual household members.

little. Conditional on employment, the estimates are therefore consistent with a compositional shift from agriculture toward wage employment among men and from flexible toward wage employment among women, although the separate employment-share regressions do not track individual transitions between job types.

[Insert Table 7 about here]

We next examine whether these employment changes translate into higher household income. To do so, we construct a household-level panel and retain the same set of controls as in the baseline specifications. Panel A of Table 8 reports the results.[26] Total household income rises by 0.073 IHS units, but the estimate does not differ detectably from zero. Non-agricultural income rises significantly by 0.046 IHS units, whereas agricultural income changes little. This pattern is consistent with a reallocation of household earnings toward non-agricultural activities and with related evidence on land rights and rural income (Hu et al., 2026; Li et al., 2026a). Exploiting the panel dimension of the data, we further split households by whether they participated in agriculture before land titling. Among households engaged in agriculture prior to the reform, total income rises significantly by 0.140 IHS units. By contrast, the corresponding estimate for households without pre-reform agricultural participation is 0.019 and does not differ detectably from zero. These results suggest that the income gains associated with land titling are concentrated among households with greater scope to reallocate labor out of agriculture.

6.2 Consumption and Household Risk Management

The employment responses above suggest that households do not simply move both spouses into the highest-earning activity. The gender-specific responses in Table 7 instead reveal an asymmetric allocation. Men's aggregate non-agricultural employment rises, and the employed subsample shows less agricultural work and more wage employment. Women's aggregate non-agricultural employment remains largely unchanged, but their conditional employment composition shows more wage employment and less flexible employment. Because flexible work generally lacks a formal labor contract, this pattern is consistent with households combining access to higher-earning wage employment with lower exposure to precarious work. The survey does not directly measure job-loss risk, however, so the risk interpretation follows from the contractual distinction between the categories rather than an estimated change in employment stability.

If such employment choices partly reflect risk management, the relevant outcome is not only household income but also how households adjust consumption in response to changes in income

[26] Each income measure is winsorized at the 1st and 99th percentiles and then transformed using the inverse hyperbolic sine (IHS).

opportunities and risk exposure. Consumption provides a natural measure of household welfare smoothing because households facing uncertain income may save part of an income gain rather than immediately increase expenditures. Related evidence from Vietnam shows that improved property rights can affect both the level and volatility of consumption, with effects that differ across households (Kemper et al., 2015).

Panel B of Table 8 therefore examines the effect of land titling on household consumption. Columns (1), (4), and (5) report total household consumption, while columns (2) and (3) separately examine infrequent and routine expenditures.[27] Across specifications, the estimated effects on consumption are small and do not differ detectably from zero. The point estimate is slightly negative among households that participated in agriculture before the reform, but it is imprecisely estimated. Thus, the increase in non-agricultural income documented in Panel A is not accompanied by a corresponding increase in current consumption. This pattern is consistent with households retaining part of the additional income rather than immediately translating it into higher expenditures, although the consumption estimates alone do not identify the underlying motive.

Appendix Table B3 provides additional evidence on short-run welfare outcomes. We find no detectable effects on self-reported health or subjective well-being. Taken together, the results indicate that land titling changes the composition of household labor and income while generating little detectable change in contemporaneous consumption or other short-run welfare measures.

[Insert Table 8 about here]

Taken together, men's aggregate response is accompanied by a conditional shift from agriculture toward wage employment. Women's aggregate non-agricultural employment remains comparatively stable, while their employment composition conditional on working shows more wage employment and less flexible employment. These changes coincide with higher non-agricultural income but no detectable increase in current consumption. The joint pattern is consistent with households assigning the larger sectoral adjustment to men while maintaining a relatively stable employment base among women; the female job-type estimates also point toward less exposure to work without formal contracts. This risk-management interpretation remains suggestive because the data do not directly measure employment risk or identify deliberate insurance through spouses' labor allocation.

[27] Infrequent expenditures include annual spending on clothing, home repairs and improvements, durable goods, education, vehicle purchases, travel, and health care. Routine expenditures include reported monthly spending on food, utilities and fuel, household necessities, local transportation, communication, culture and entertainment, domestic services, and personal care.

7 Conclusion

Land institutions that tie tenure security to continued cultivation create a friction: households that might otherwise reallocate labor to more productive non-agricultural activities must weigh the risk of losing their land against the gains from mobility. This friction operates within the household because members differ in their access to jobs and their household responsibilities. Drawing on village-level titling completion years and four waves of the China Household Finance Survey, we document a sharp gender contrast following China's land-titling reform: men's non-agricultural employment rises, while women's aggregate non-agricultural employment does not. Among employed women, however, wage employment rises while flexible employment falls. Together with higher non-agricultural income and no detectable increase in current consumption, these patterns are consistent with a household-risk-management interpretation in which men undertake the larger sectoral shift while women's overall cross-sector employment allocation remains comparatively stable; within the non-agricultural sector, the job-type estimates point toward more wage work and less work without formal contracts.

The broader implication is that removing a land-related mobility friction does not guarantee a uniform response within the household. Evaluations of land-rights reform should therefore report employment margins separately across household members rather than infer shared labor-market gains from a household-level outcome. The larger male coefficients in places with more accessible jobs and lighter observed care demands also point to possible complementarities with labor-market access and care provision. This paper does not estimate the effects of those policies, but it suggests that the gains from land titling depend on the economic and household setting in which those rights are used. Securing a household's land claim can expand mobility while households distribute the resulting adjustment unevenly across spouses. In our setting, the larger shift among men and the relative stability of women's overall sectoral allocation provide a coherent account of how households balance higher expected returns against income risk.

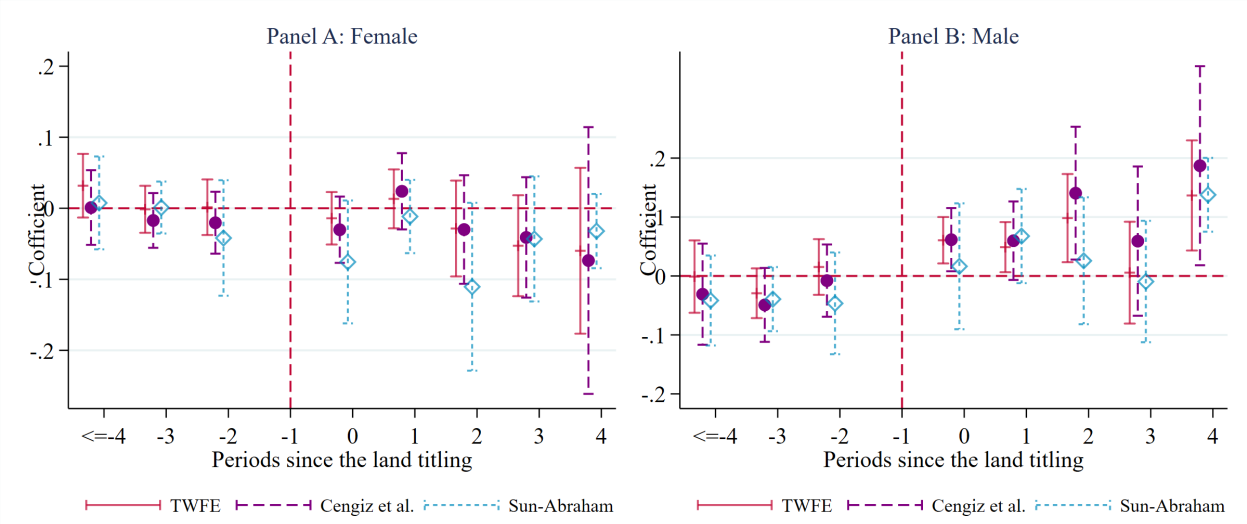


Fig. 1. Dynamic Effects by Gender

Note: The figure plots event-study coefficients and 95 percent confidence intervals for non-agricultural employment. Panel A reports estimates for women, and Panel B reports estimates for men. The omitted category is the village completion year ($k = -1$); $k = 0$ denotes the calendar year immediately after completion. The tails are grouped at $k \leq -4$ and $k \geq 4$. Solid, dashed, and dotted series correspond to the two-way fixed-effects estimator, the stacked DID estimator of Cengiz et al. (2019), and the interaction-weighted estimator of Sun and Abraham (2021), respectively. Because the survey is not conducted every calendar year, $k = 0$ need not be an observed survey period for every village. Error bars show 95 percent confidence intervals, with standard errors clustered at the village level. Data are from the CHFS.

[Back to Page 9]

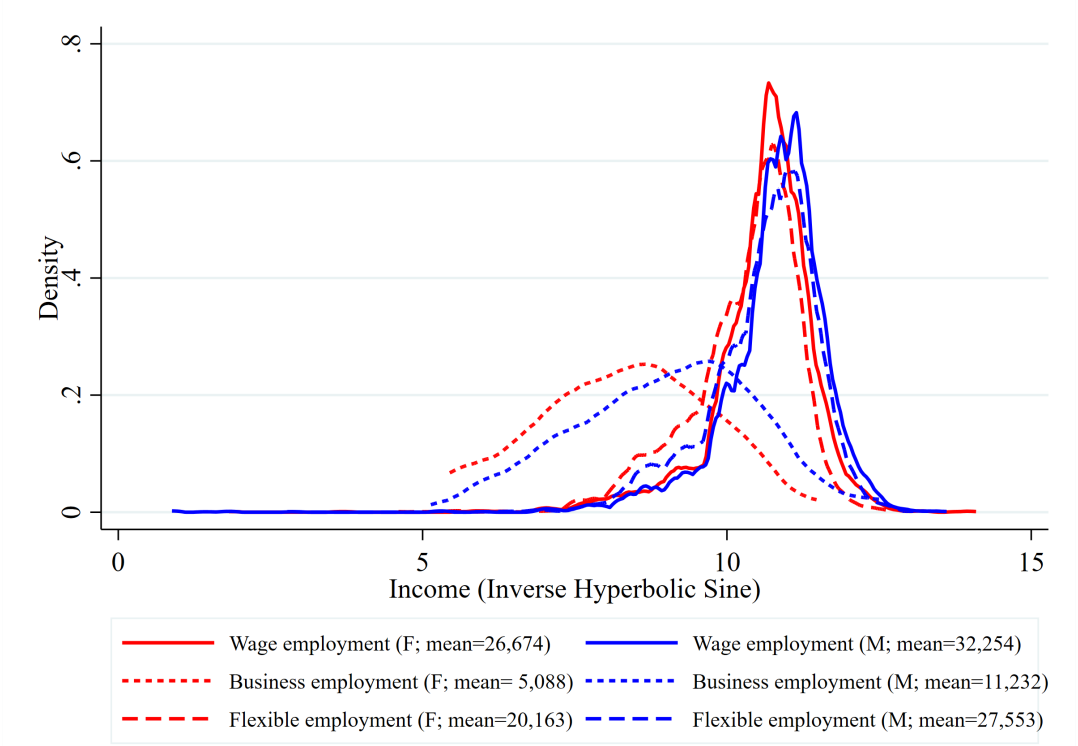


Fig. 2. Income by Non-agricultural Job Type and Gender

Notes: The figure plots kernel density estimates of individual income by primary non-agricultural job type among respondents with positive reported income. Blue lines denote men and red lines denote women. Solid, dotted, and long-dashed lines denote wage employment, business, and flexible employment, respectively. Following Burbidge et al. (1988), income is transformed using the inverse hyperbolic sine (IHS), $\text{asinh}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$, which is defined at zero and approximates the natural logarithm at higher values. The legend uses the same color and line conventions and reports the corresponding group means in yuan. The densities are descriptive, do not condition on covariates, and do not estimate the effect of land titling. Data are from the CHFS.

[Back to Page 17]

Table 1. Summary Statistics

	Obs.	Min	Mean	Max	Std. Dev.
Panel A: Village Characteristics					
Population	1,507	110.00	1,872.50	18,352.00	1,640.75
Out-Migrants	1,500	0.00	440.97	2,530.00	478.85
Large Surname Group (Yes=1)	1,568	0.00	0.95	1.00	-
One-Shoulder Arrangement (Yes=1)	1,275	0.00	0.37	1.00	-
Village Area (1,000 *mu*)	1,565	0.15	95.81	9,000	749.37
Distance to County Center (km)	1,372	0.00	30.97	140.00	25.08
Panel B: Household Characteristics					
Household Size	24,680	1.00	4.28	19.00	1.72
Per Capita Income (1,000 yuan)	23,859	0.00	13.14	311.10	17.71
Total Consumption (1,000 yuan)	24,680	0.41	12.30	217.79	14.14
Total Assets (1,000 yuan)	24,680	0.15	104.68	5,152.65	224.39
Total Debt (1,000 yuan)	18,601	0.00	11.20	585.00	29.53
Total Savings (1,000 yuan)	24,680	0.00	4.48	305.00	14.80
Panel C: Individual Characteristics					
Age	69,829	17.00	39.98	60.00	12.49
Years of Schooling	69,829	0.00	8.70	22.00	4.24
Employed	69,829	0.00	0.77	1.00	-
Agri. Employment	69,829	0.00	0.35	1.00	-
Non-agri. Employment	69,829	0.00	0.40	1.00	-
Flexible Employment	69,829	0.00	0.20	1.00	-
Wage Employment	69,829	0.00	0.15	1.00	-
Business Employment	69,829	0.00	0.05	1.00	-

Notes: Data are drawn from the CHFS. Continuous economic variables at the village and household levels are winsorized at the 1st and 99th percentiles. Out-migrants are members of the village labor force working outside their place of household registration or usual household residence. The large-surname-group indicator (*Daxing*) equals one when a single surname accounts for more than 10 percent of the village population and proxies for a strong kinship-based structure. The one-shoulder arrangement (*yijiantiao*) places the village Party branch secretary and village committee director positions under the same person. Flexible employment includes temporary work without a formal labor contract, such as casual work, as well as freelance and volunteer work. One *mu* equals 1/15 of a hectare. Years of schooling are coded as follows: 0 = no schooling or incomplete primary, 6 = primary school, 9 = junior high school, 12 = senior high, vocational, or technical school, 16 = college or university, and 19 = graduate education or above. Economic variables are measured in 1,000 yuan.

[Back to Page 7]

Table 2. Effect on Non-agri. Employment

	Outcome: Non-agri. Employment			
	All	Female	Male	Pooled Interaction
	(1)	(2)	(3)	(4)
Treated×Post	0.024**	-0.002	0.045***	0.038***
	(0.010)	(0.014)	(0.015)	(0.011)
Treated×Post×Female				-0.027***
				(0.010)
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes
Observations	69,829	30,660	35,554	69,829
Adj. R^2	0.212	0.261	0.257	0.260

Notes: This table presents the effect of land titling on non-agricultural employment. The dependent variable equals one when the respondent's primary occupation is outside agriculture and zero when the respondent works in agriculture or is not employed. Columns (1)–(3) report estimates for the full, female, and male samples, respectively. Column (4) uses the full sample and reports the coefficients on the land-titling indicator and its interaction with the female indicator. In this pooled specification, the coefficient on Treated×Post is the treatment response for men, the omitted gender category, and the interaction coefficient measures the female treatment response relative to the male response. All regressions include household and county-by-year fixed effects and individual controls for age and years of schooling; the pooled specifications additionally control for gender. No additional household-level covariates are included. The gender-specific samples can lose observations because the fixed-effects estimator omits singleton household groups, so their observation counts need not sum to the pooled-sample count. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 8]

Table 3. Robustness Checks

	Outcome: Non-agri. Employment					
Panel A: Alternative Sample and Fixed Effects						
	Exclude Sichuan			Individual FE		
	(1) Female	(2) Male	(3) Pooled	(4) Female	(5) Male	(6) Pooled
Treated×Post	-0.003	0.050***	0.040***	-0.010	0.029*	0.025*
	(0.014)	(0.015)	(0.011)	(0.016)	(0.017)	(0.014)
Treated×Post×Female			-0.028***			-0.032***
			(0.010)			(0.012)
Observations	29,156	32,009	66,483	25,921	28,133	54,121
Adj. R^2	0.260	0.257	0.259	0.444	0.427	0.459
Panel B: More controls and Prior Occupation						
	More controls			Prior Farmers		
	(1) Female	(2) Male	(3) Pooled	(4) Female	(5) Male	(6) Pooled
Treated×Post	0.007	0.040**	0.044***	0.032	0.116***	0.125***
	(0.016)	(0.016)	(0.013)	(0.025)	(0.037)	(0.025)
Treated×Post×Female			-0.032***			-0.096***
			(0.010)			(0.017)
Observations	26,938	29,425	60,012	8,387	7,599	16,843
Adj. R^2	0.257	0.258	0.259	0.182	0.225	0.204
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is non-agricultural employment. Within each specification, the first two columns report female- and male-subsample estimates, and the third reports the pooled model's land-titling coefficient and its interaction with the female indicator. In the pooled columns, Treated×Post is the treatment response for men, the omitted gender category, and the interaction coefficient is the female response relative to the male response. Panel A excludes Sichuan Province in columns (1)–(3) and replaces household with individual fixed effects in columns (4)–(6). The five 2013 village characteristics interacted with year indicators in Panel B are out-migrants, a large surname group (*Daxing*), distance to the county center, registered (*hukou*) population, and farmland per capita. Registered population counts village *hukou* holders; out-migrants are village workers employed outside their place of household registration or usual residence, with missing 2013 counts replaced by 2015 reports. Columns (4)–(6) in Panel B restrict the sample to individuals working in agriculture before their village completed titling. Gender-specific regressions can omit additional singleton household groups, so their observation counts need not sum to the pooled count. All regressions include the indicated fixed effects and controls for age and years of schooling; pooled models also control for gender. Standard errors are clustered at the village level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 10]

Table 4. Heterogeneity: Household Structure

	Female		Male	
	(1)	(2)	(3)	(4)
Panel A: Care Burden—Children				
	With	Without	With	Without
Land Titling	-0.011	-0.002	0.018	0.060*
	(0.022)	(0.026)	(0.031)	(0.033)
Observations	14,167	8,773	13,314	8,268
Adj. R^2	0.325	0.420	0.361	0.429
Panel B: Care Burden—Unhealthy Older Members				
	With	Without	With	Without
Land Titling	-0.006	-0.012	-0.001	0.046**
	(0.056)	(0.017)	(0.075)	(0.020)
Observations	2,696	19,820	2,389	18,778
Adj. R^2	0.328	0.355	0.391	0.404
Panel C: Marital Status				
	Married	Other	Married	Other
Land Titling	-0.009	-0.083	0.036*	0.050
	(0.015)	(0.059)	(0.020)	(0.036)
Observations	24,160	4,517	22,756	7,721
Adj. R^2	0.282	0.294	0.294	0.334
Panel D: Relative Education				
	High	Low	High	Low
Land Titling	-0.016	-0.010	0.004	0.084**
	(0.025)	(0.031)	(0.034)	(0.038)
Observations	8,698	5,927	8,159	5,024
Adj. R^2	0.496	0.469	0.454	0.476
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes

Notes: The dependent variable is non-agricultural employment. Panels A and B restrict the sample to married respondents and split households by the presence of a child aged 15 or below and an unhealthy member aged 60 or above, respectively. Health ranges from 1 (very good) to 5 (very poor); the unhealthy indicator equals one for responses of 4 or 5. Panel C splits the sample by marital status. In Panel D, "High" denotes couples in which the wife's schooling equals or exceeds the husband's, and "Low" denotes couples in which the wife has less schooling. Columns (1) and (2) report estimates for women; columns (3) and (4) report estimates for men. All regressions include household and county-by-year fixed effects and controls for age and years of schooling. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 13]

Table 5. Heterogeneity: Local Labor Markets

	Female		Male	
	(1)	(2)	(3)	(4)
Panel A: Occupational Segregation				
	High	Low	High	Low
Land Titling	-0.009	0.013	0.066***	0.028
	(0.017)	(0.022)	(0.019)	(0.024)
Observations	15,518	14,157	17,163	15,383
Adj. R^2	0.251	0.269	0.258	0.259
Panel B: Industrial Firms				
	More	Less	More	Less
Land Titling	0.002	0.002	0.056***	0.026
	(0.019)	(0.021)	(0.021)	(0.023)
Observations	17,769	12,419	19,411	13,710
Adj. R^2	0.257	0.237	0.244	0.255
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes

Notes: This table reports heterogeneity by local labor-market structure. Panel A splits counties at the median Duncan occupational segregation index, constructed from the 2010 Population Census of China. Panel B splits counties at the median number of industrial enterprises above the designated size reported in county statistical yearbooks. These firms have annual revenue from principal business of at least 20 million yuan. Columns (1) and (2) report estimates for women; columns (3) and (4) report estimates for men. Within each gender, the first column is the above-median group and the second is the below-median group. All regressions include household and county-by-year fixed effects and individual controls for age and years of schooling. No additional household-level covariates are included. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 14]

Table 6. Migration and Geographic Mobility

	(1)	(2)	(3)	(4)
Panel A: Migration				
	Migration		Non-agri. Employment	
	Female	Male	Female	Male
Land Titling	0.015	0.020	-0.007	0.039***
	(0.011)	(0.014)	(0.014)	(0.015)
Migration			0.324***	0.294***
			(0.015)	(0.013)
Observations	30,660	35,554	30,660	35,554
Adj. R^2	0.201	0.233	0.297	0.289
Panel B: Distance to County Center				
	Non-agri. Employment: Far		Non-agri. Employment: Close	
	Female	Male	Female	Male
Land Titling	0.024	0.038*	-0.034	0.109***
	(0.018)	(0.020)	(0.028)	(0.035)
Observations	17,351	19,160	13,230	14,313
Adj. R^2	0.248	0.265	0.258	0.232
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes

Notes: Panel A uses the baseline sample. Columns (1) and (2) report the effect of land titling on migration, while columns (3) and (4) report effects on non-agricultural employment after controlling for migration status. A household member is defined as a migrant if they leave the county for more than three months. Panel B reports heterogeneous effects on non-agricultural employment by distance to the county center. "Close" villages lie below the sample median distance of approximately 31 kilometers; "Far" villages comprise the remaining observations. All regressions include household and county-by-year fixed effects and individual controls for age and years of schooling; no additional household-level covariates are included. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 14]

Table 7. Effect on Employment Margins

	(1) Female	(2) Male	(3) Female	(4) Male	(5) Female	(6) Male	(7) Female	(8) Male
	Agriculture		Wage		Flexible		Business	
Land Titling	0.011	−0.047***	0.029*	0.028**	-0.032**	0.001	-0.007	0.002
	(0.018)	(0.016)	(0.015)	(0.014)	(0.016)	(0.016)	(0.012)	(0.009)
Observations	18,938	26,237	18,938	26,237	18,938	26,237	18,938	26,237
Adj. R^2	0.561	0.477	0.361	0.347	0.264	0.295	0.325	0.326
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample is restricted to respondents who are employed in the survey year, so the estimates describe changes in primary-employment composition conditional on employment. Columns (1) and (2) report agricultural employment, while columns (3)–(8) report wage employment, flexible employment, and business employment, including self-employment. Flexible employment includes temporary work without a formal labor contract, such as casual work, as well as freelance and volunteer work. Columns (1), (3), (5), and (7) report estimates for women; columns (2), (4), (6), and (8) report estimates for men. All regressions include household and county-by-year fixed effects and individual controls for age and years of schooling. No additional household-level covariates are included. Standard errors clustered at the village level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 17]

Table 8. Effect on Household Income and Consumption

	(1) Total	(2) Agri.	(3) Non-agri.	(4) Pre-reform agri HH	(5) Pre-reform Non-agri HH
Panel A: Income					
Land Titling	0.073	-0.011	0.046*	0.140*	0.019
	(0.062)	(0.210)	(0.026)	(0.080)	(0.106)
Observations	19,692	19,020	20,537	10,295	9,352
Adj. R^2	0.277	0.485	0.964	0.289	0.259
Panel B: Consumption					
	(1) Total	(2) Infrequent	(3) Routine	(4) Pre-reform agri HH	(5) Pre-reform Non-agri HH
Land Titling	-0.014	0.083	-0.001	-0.024	-0.039
	(0.036)	(0.096)	(0.033)	(0.058)	(0.044)
Observations	20,537	20,537	20,537	8,701	11,794
Adj. R^2	0.425	0.268	0.486	0.389	0.513
Controls	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the effect of land titling on annual household income and consumption. The dependent variables in Panel A are total household income in columns (1), (4), and (5), agricultural income in column (2), and non-agricultural income in column (3). The dependent variables in Panel B are total household consumption in columns (1), (4), and (5), infrequent expenditures in column (2), and routine expenditures in column (3). Each underlying income and consumption measure is winsorized at the 1st and 99th percentiles and then transformed using the inverse hyperbolic sine, $\text{asinh}(y) = \ln\left(y + \sqrt{y^2 + 1}\right)$. Columns (1)–(3) use the full sample. Column (4) restricts the sample to households with at least one pre-reform agricultural worker, while column (5) restricts the sample to households with no observed pre-reform agricultural worker. All regressions include household and county-by-year fixed effects and the baseline control variables described in Section 3. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 18]

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A Figures

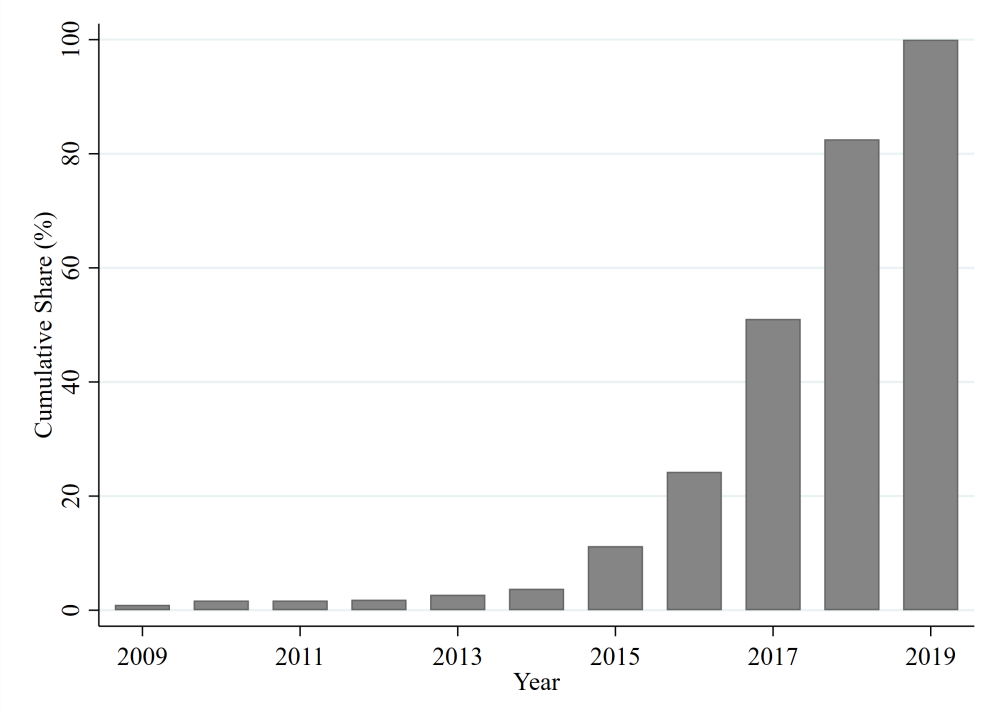


Fig. A1. Implementation Progress of Land Titling

Notes: This figure plots the cumulative share of villages that had completed land titling by each year. The denominator includes villages with reported completion years. Data are from the CHFS.

[Back to Page 4]

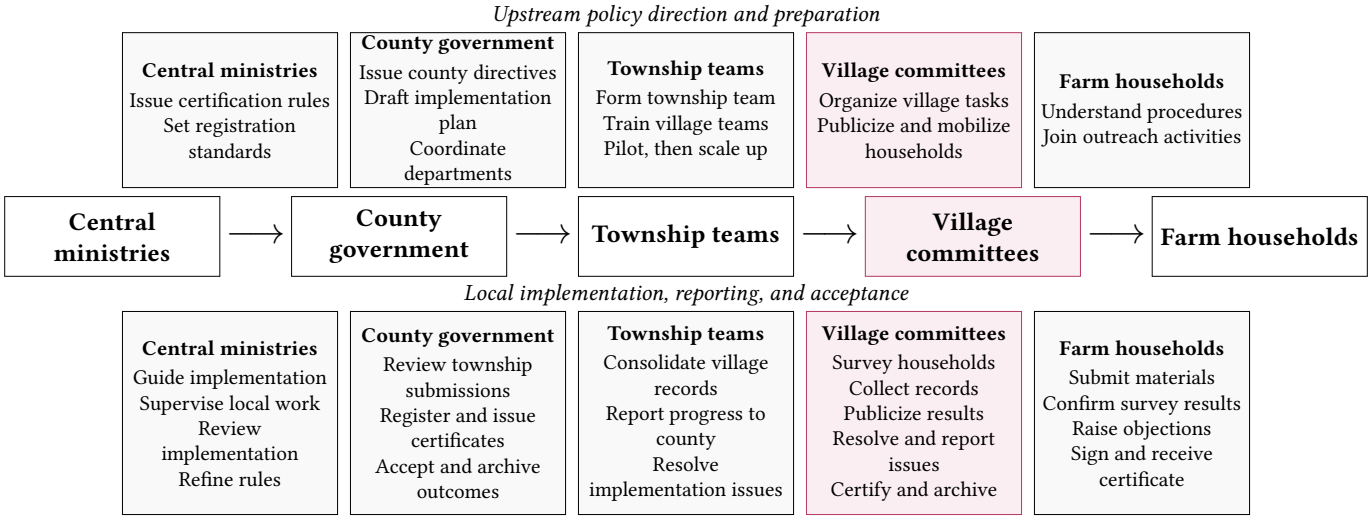


Fig. A2. Administrative Chain of Land Titling Reform

Notes: The diagram summarizes the administrative chain of land titling. The upper boxes list directives from higher levels of government, while the lower boxes list implementation, reporting, review, and household-confirmation tasks. Arrows indicate the broad deployment and feedback sequence. County governments coordinate the process, township teams organize and supervise village work, village committees conduct the local survey and certification, and farm households provide materials, verify records, and receive certificates. Because these steps were completed locally and not simultaneously across villages, the empirical treatment is assigned using the village completion year rather than a common county launch date. The listed tasks follow the national implementation guidance and archival measures cited in the text, including records review, parcel survey, household confirmation, register updates, certificate issuance, and archiving.

[Back to Page 4]

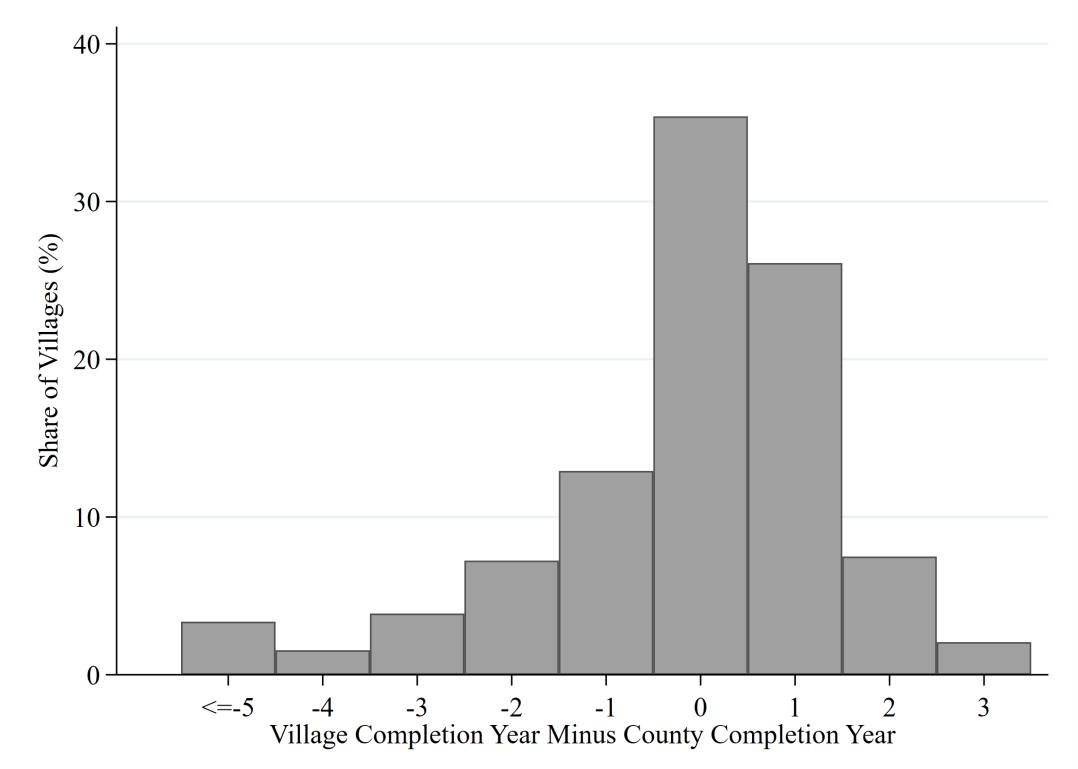


Fig. A3. Within-County Variation in Completion Timing

Notes: This figure plots the share of villages at each difference between the village and county completion years, calculated as the village year minus the county year. The sample includes villages for which both completion years are observed. Zero indicates completion in the same calendar year. Negative values indicate that a village completed titling before the corresponding county completion year, whereas positive values indicate that it completed titling afterward.

[Back to Page 6]

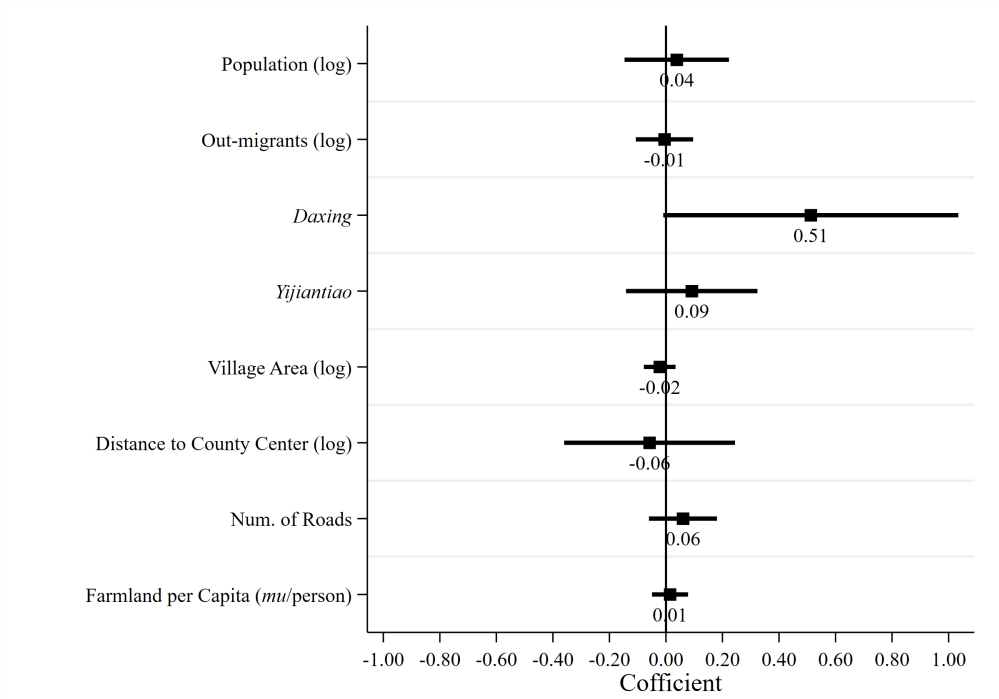


Fig. A4. Exogeneity of Treatment Timing

Notes: This figure examines whether survey-baseline village characteristics predict the timing of land titling. Each marker reports the coefficient from a separate regression of the village's titling-completion year on one characteristic. The characteristics include village population, out-migration, the presence of a large surname group (*Daxing*), the *yijiantiao* ("one-shoulder") arrangement, village area, distance to the county center, the number of roads, and farmland per capita. Out-migrants are members of the village labor force working outside their place of household registration or usual household residence. Error bars show 95 percent confidence intervals, with standard errors clustered at the village level.

[Back to Page 10]

B Tables

This section collects the supplementary tables referenced in the text.

Table B1. Alternative Staggered-Adoption Estimates

	Average effect on non-agri. employment		
	TWFE	Sun–Abraham	Stacked DID
Female	-0.002	-0.008	-0.020
	(0.014)	(0.019)	(0.033)
Male	0.045***	0.057***	0.091**
	(0.015)	(0.020)	(0.042)

Notes: This table reports gender-specific average post-treatment effects from two alternative staggered-adoption estimators proposed by Sun and Abraham (2021) and Cengiz et al. (2019), with TWFE shown for comparison. The Sun–Abraham interaction-weighted estimator and the stacked DID estimator use all pre-treatment periods as control groups and report an equal-weight average of the post-treatment coefficients. All specifications include household and county-by-year fixed effects and individual controls for age and years of schooling; no additional household-level covariates are included. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 9]

Table B2. Heterogeneity: Joint Care Demands

	With Children		Without Children	
Panel A: With Unhealthy Elderly				
	Female	Male	Female	Male
Land Titling	0.090	-0.027	0.069	0.052
	(0.097)	(0.086)	(0.123)	(0.227)
Observations	1,693	1,422	588	568
Adj. R^2	0.289	0.364	0.429	0.386
Panel B: Without Unhealthy Elderly				
	With Children		Without Children	
	Female	Male	Female	Male
Land Titling	-0.034	0.009	-0.008	0.082**
	(0.028)	(0.033)	(0.029)	(0.033)
Observations	11,528	10,838	7,243	6,918
Adj. R^2	0.317	0.369	0.423	0.445
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes

Notes: This table reports the joint heterogeneity analysis of family care demands. The sample is restricted to married individuals, and the dependent variable is non-agricultural employment. Panel A restricts the sample to households with an unhealthy member aged 60 or above and then splits it by the presence of children aged 15 or below. Panel B applies the same child split to households without an unhealthy older member. The underlying health measure ranges from 1 (very good) to 5 (very poor); the unhealthy indicator equals one for poor or very poor health (4 or 5) and zero otherwise. All regressions include household and county-by-year fixed effects and individual controls for age and years of schooling. Standard errors clustered at the village level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 12]

Table B3. Effect on Subjective Well-Being

	Health		Happiness	
	Female	Male	Female	Male
Land Titling	0.002	-0.012	0.037	0.038
	(0.019)	(0.020)	(0.026)	(0.026)
Controls	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
County-by-Year FE	Yes	Yes	Yes	Yes
Observations	25,855	27,754	30,630	33,522
Adj. R^2	0.390	0.375	0.422	0.437

Notes: This table reports the effect of land titling on self-reported health and subjective happiness by gender. The health question asks respondents to assess their current physical condition relative to people of the same age. The original health and happiness scales range from 1 to 5, with higher values indicating better health or greater happiness. For both outcomes, responses from 1 to 3 are coded as zero and responses of 4 or 5 are coded as one. Columns (1) and (2) report effects on health for women and men, respectively; columns (3) and (4) report effects on happiness. All regressions include household and county-by-year fixed effects and individual controls for age and years of schooling. Standard errors clustered at the village level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[Back to Page 19]